

Domain Name System (DNS)

- Topics covered:
 - The Purpose of DNS
 - Basic Functions of DNS
 - Configuring DNS in Cisco IOS

The Purpose of DNS

Introduction to DNS

DNS is used to resolve (convert) human-readable names into IP addresses.

- Example: youtube.com → converted to an IP address
- Devices (PCs, routers) use IP addresses, not names
- Humans prefer names because they are easier to remember

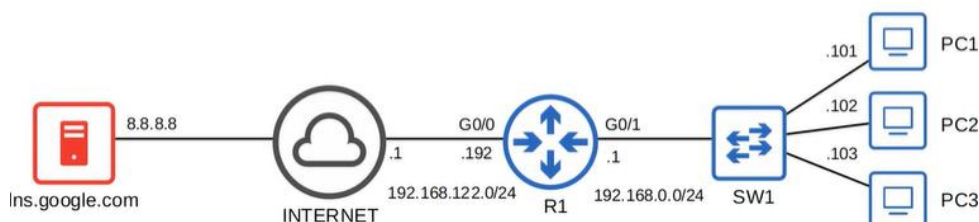
Even though IPv4 addresses are readable, names are still much easier to use. Without DNS, users would need to remember IP addresses for every website.

When a name is entered (e.g., in a browser), the device sends a request to a **DNS server**, which replies with the correct IP address.

DNS servers can be:

- Manually configured
- Learned automatically via DHCP

Network Used for Demonstration



- PCs connect to R1 via SW1
- R1 connects to the Internet
- DNS server used: 8.8.8.8 (Google DNS)

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Checking DNS Configuration on PC

CLI of PC1

```
C:\Users\user>ipconfig /all

[output omitted]

Ethernet adapter ローカル エリア接続 :

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) 82579LM Gigabit Network Connection
    Physical Address. . . . . : 78-2B-CB-AC-08-67
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.0.101(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
    DNS Servers . . . . . : 8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled

[output omitted]
```

Explanation:

The command `ipconfig /all` displays full network details.

- It shows the configured **DNS server (8.8.8.8)**
- This is the server the PC will use to resolve names

Note:

- `ipconfig` (without `/all`) does not show this detail
- Normally DNS is learned via DHCP, but here it is manually configured

Resolving Names using NSLOOKUP

CLI Output of PC1:

```
C:\Users\user>nslookup youtube.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     youtube.com
Addresses: 2404:6800:4004:819::200e
          172.217.25.110
```

Explanation:

- `nslookup` = **Name Server Lookup**
- It asks the DNS server for the IP of a domain

Result:

- IPv4 address → 172.217.25.110
- IPv6 address → also returned

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Using the Name Directly (Ping)

CLI Output of PC1:

```
C:\Users\user>ping youtube.com

Pinging youtube.com [172.217.25.110] with 32 bytes of data:
Reply from 172.217.25.110: bytes=32 time=10ms TTL=117
Reply from 172.217.25.110: bytes=32 time=7ms TTL=117
Reply from 172.217.25.110: bytes=32 time=7ms TTL=117
Reply from 172.217.25.110: bytes=32 time=7ms TTL=117

Ping statistics for 172.217.25.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 10ms, Average = 7ms
```

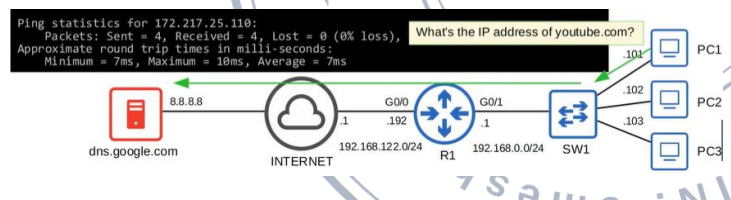
Explanation:

- The device automatically resolves the name before sending packets
- No need to run nslookup first
- DNS resolution happens in the background

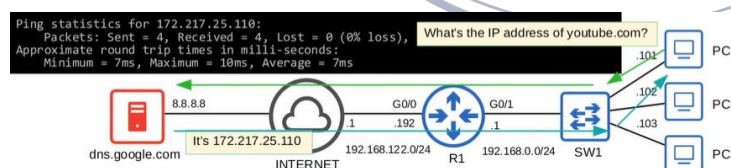
*you don't have to use the **nslookup** command before sending the ping. If your device doesn't know the correct IP address it will automatically ask the server.

How DNS Works (Step-by-Step Flow)

1. PC1 → DNS Query → 8.8.8.8



2. DNS Server → Response → 172.217.25.110



Explanation:

- PC1 sends a **DNS query** asking for the IP of youtube.com
- DNS server replies with the IP address

Important Role of R1:

1. R1 is not a DNS server or client here
2. It only forwards packets between PC1 and DNS server
3. Therefore: No DNS configuration is required on R1

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DNS Messages (Wireshark Capture)

No.	Time	Source	Destination	Protocol	Length	Info
1087	08:55:44.458619	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0002 A youtube.com
1088	08:55:44.508043	8.8.8.8	192.168.0.101	DNS	87	Standard query response 0x0002 A youtube.com A 172.217.25.110
1089	08:55:44.508888	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0003 AAAA youtube.com
1115	08:55:44.641775	8.8.8.8	192.168.0.101	DNS	99	Standard query response 0x0003 AAAA youtube.com AAAA 2404:6800:4004:819::200e

> Frame 1087: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF_{9956EC07-3774-4811-9780-C823E7CD172}, id 0
> Ethernet II, Src: Dell_ac:08:07 (78:2b:cb:ac:08:07), Dst: Tp-LinkT_dd:a8:e4 (98:da:c4:dd:a8:e4)
> Internet Protocol Version 4, Src: 192.168.0.101, Dst: 8.8.8.8
> User Datagram Protocol, Src Port: 49286, Dst Port: 53
▼ Domain Name System (query)
Transaction ID: 0x0002
▼ Flags: 0x0100 Standard query
0... .. = Response: Message is a query
0000... .. = Opcode: Standard query (0)
... ..0... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..0... .. = Z: reserved (0)
... ..0... .. = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
▼ Queries
▼ youtube.com: type A, class IN
Name: youtube.com
[Name Length: 11]
[Label Count: 2]
Type: A (Host Address) (1)
Class: IN (0x0001)
[Response In: 1088]

Query 1: PC1 → DNS → A youtube.com

Reply 1: DNS → PC1 → IPv4 address

Query 2: PC1 → DNS → AAAA youtube.com

Reply 2: DNS → PC1 → IPv6 address

Explanation:

Two types of DNS records are shown:

1. A record → maps name to IPv4 address
2. AAAA record → maps name to IPv6 address

No.	Time	Source	Destination	Protocol	Length	Info
1087	08:55:44.458619	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0002 A youtube.com
1088	08:55:44.508043	8.8.8.8	192.168.0.101	DNS	87	Standard query response 0x0002 A youtube.com A 172.217.25.110
1089	08:55:44.508888	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0003 AAAA youtube.com
1115	08:55:44.641775	8.8.8.8	192.168.0.101	DNS	99	Standard query response 0x0003 AAAA youtube.com AAAA 2404:6800:4004:819::200e

Query 1 More Detail

> Frame 1087: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface \Device\NPF_{9956EC07-3774-4811-9780-C823E7CD172}, id 0 > Ethernet II, Src: Dell_ac:08:07 (78:2b:cb:ac:08:07), Dst: Tp-LinkT_dd:a8:e4 (98:da:c4:dd:a8:e4) > Internet Protocol Version 4, Src: 192.168.0.101, Dst: 8.8.8.8 > User Datagram Protocol, Src Port: 49286, Dst Port: 53 ▼ Domain Name System (query) Transaction ID: 0x0002 ▼ Flags: 0x0100 Standard query 0... .. = Response: Message is a query 0000... .. = Opcode: Standard query (0)0... .. = Truncated: Message is not truncated1... .. = Recursion desired: Do query recursively0... .. = Z: reserved (0)0... .. = Non-authenticated data: Unacceptable Questions: 1 Answer RRs: 0 Authority RRs: 0 Additional RRs: 0 ▼ Queries ▼ youtube.com: type A, class IN Name: youtube.com [Name Length: 11] [Label Count: 2] Type: A (Host Address) (1) Class: IN (0x0001) [Response In: 1088]
--

Protocol Details

- DNS uses:
 - UDP (usually)
 - TCP (if message > 512 bytes)
- Port number: 53

Standard queries like these typically use UDP.

Basic Functions of DNS

DNS Cache

Devices store DNS responses locally to avoid repeated queries.

CLI Output of PC:

```
C:\Users\user>ipconfig /displaydns
[output omitted]
www.youtube.com
-----
Record Name . . . . . : www.youtube.com
Record Type . . . . . : 5
Time To Live . . . . . : 98
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : youtube-ui.l.google.com
[output omitted]
Record Name . . . . . : youtube-ui.l.google.com
Record Type . . . . . : 1
Time To Live . . . . . : 98
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . . : 172.217.25.110
[output omitted]
```

Explanation:

- DNS cache stores previous results
- This avoids querying DNS server every time

Important record type:

- **CNAME (Canonical Name)** → maps one name to another name
- Then resolved to IP using an **A record**

Clearing DNS Cache:

```
C:\Users\user>ipconfig /flushdns
Windows IP Configuration
Successfully flushed the DNS Resolver Cache.
C:\Users\user>ipconfig /displaydns
Windows IP Configuration
```

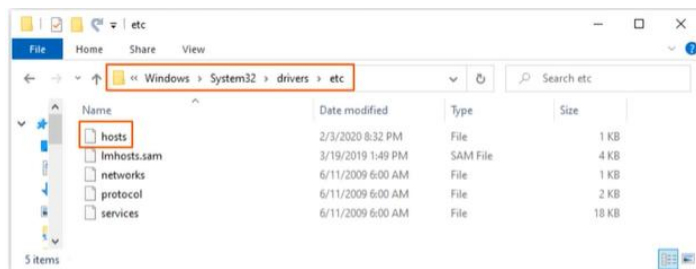
Explanation:

1. ipconfig /flushdns clears all cached records
2. After clearing:
 - Device must query DNS server again

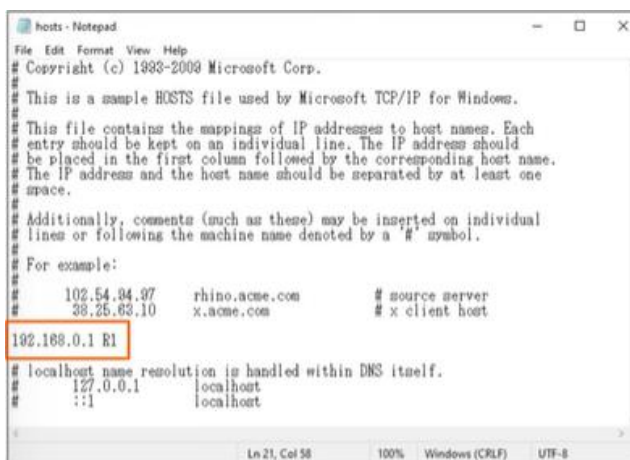
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Hosts File (Alternative to DNS)

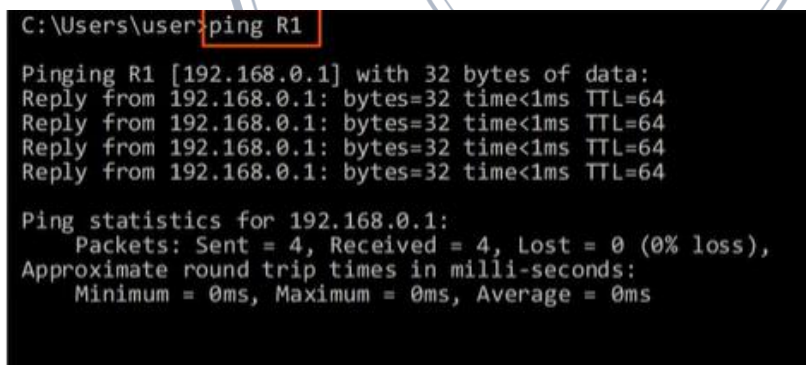
Location: “C:\Windows\System32\drivers\etc\hosts”



Example Entry: 192.168.0.1 R1



CLI Output:



Explanation:

1. Hosts file manually maps:
 - o IP address → hostname
2. No DNS query is needed if entry exists

Important context:

1. Used before DNS existed
2. Still useful for small/local setups
3. DNS is the preferred modern solution

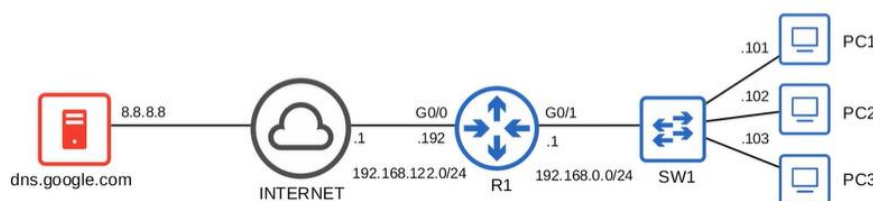
Configuring DNS in Cisco IOS

Configuring a Cisco Router as a DNS Server

Previously, DNS was used from external servers. Now the router itself (R1) is configured to **act as a DNS server**.

Routers normally just **forward DNS packets**, and no configuration is required. However:

1. A Cisco router can act as a DNS server (rare in real networks)
2. Usually, internal DNS servers are Windows/Linux servers
3. A router can also act as a DNS client to resolve names for its own use



CLI Configuration (R1 as DNS Server)

<pre>R1(config)#ip dns server</pre>	Configure R1 to act as a DNS server.
<pre>R1(config)#ip host R1 192.168.0.1 R1(config)#ip host PC1 192.168.0.101 R1(config)#ip host PC2 192.168.0.102 R1(config)#ip host PC3 192.168.0.103</pre>	Configure a list of hostname/IP address mappings.
<pre>R1(config)#ip name-server 8.8.8.8</pre>	Configure a DNS server that R1 will query if the requested record isn't in its host table.
<pre>R1(config)#ip domain lookup</pre>	Enable R1 to perform DNS queries. (enabled by default) (old version of the command is <code>ip domain-lookup</code>)

Configuration builds logically:

1. Enable DNS service on R1
2. Create **local DNS records** using `ip host`
3. Configure external DNS (8.8.8.8) for unknown queries
4. Ensure DNS lookup is enabled (`ip domain lookup`)

Notes:

- `ip domain lookup` is enabled by default
- Older command: `ip domain-lookup`

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Now R1 can:

- Answer DNS queries locally
- Query external DNS servers when needed

Local DNS Resolution (PC1 → PC2)

CLI Output of PC1:

```
C:\Users\user>ipconfig /all

[output omitted]

IPv4 Address. . . . . : 192.168.0.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.1
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled

[output omitted]
```

I configure PC1 To use DNS server as R1

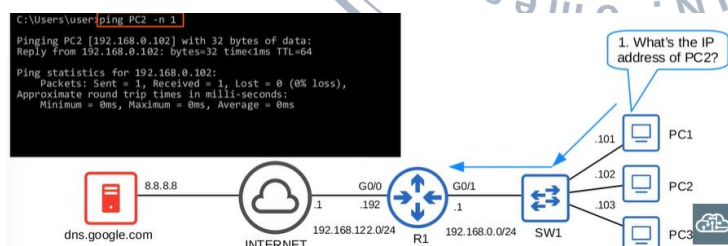
```
C:\Users\user>ping PC2 -n 1

Pinging PC2 [192.168.0.102] with 32 bytes of data:
Reply from 192.168.0.102: bytes=32 time<1ms TTL=64

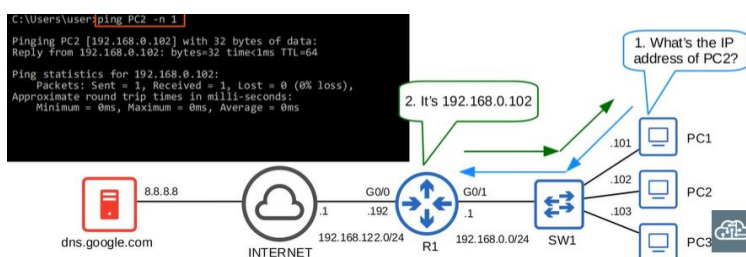
Ping statistics for 192.168.0.102:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Flow Diagram:

1. PC1 → R1 "What is the IP of PC2?"

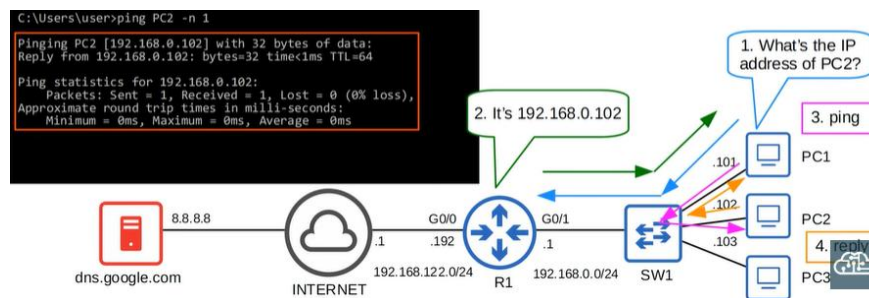


2. R1 → PC1 "192.168.0.102"



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3. PC1 → PC2 (Ping) & 4. PC2 → PC1 (Reply)



Explanation

PC1 does not know PC2's IP initially.

1. It sends a DNS query to R1
2. R1 checks its **local host table**
3. Finds entry (configured using ip host)
4. Replies with IP address

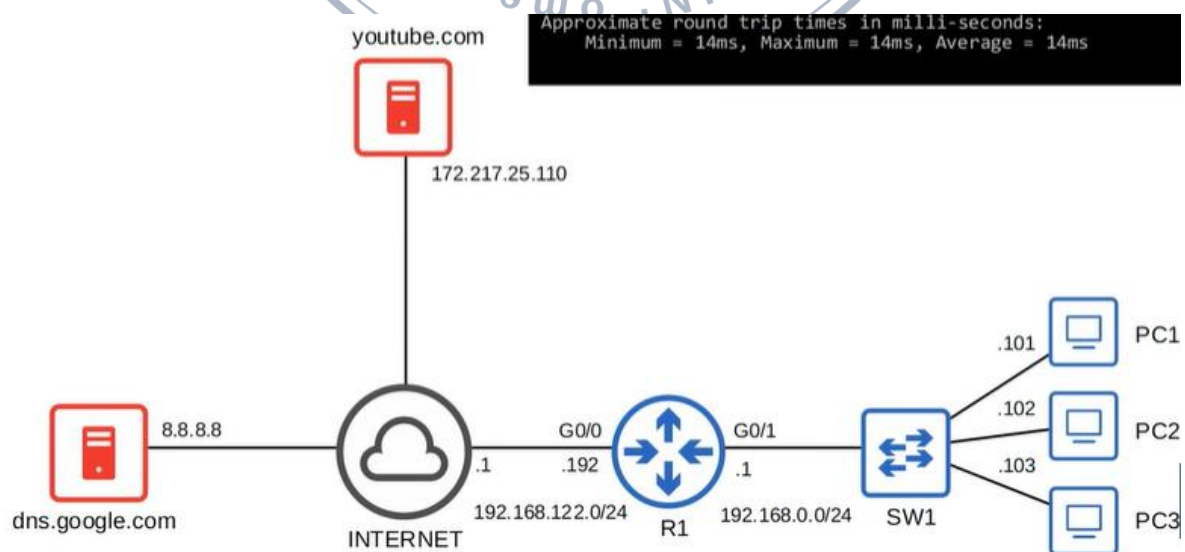
Then:

- PC1 sends ping
- PC2 replies

After this:

- PC1 stores result in **DNS cache**
- Future queries are not needed

External DNS Resolution (youtube.com)



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CLI Output of PC1:

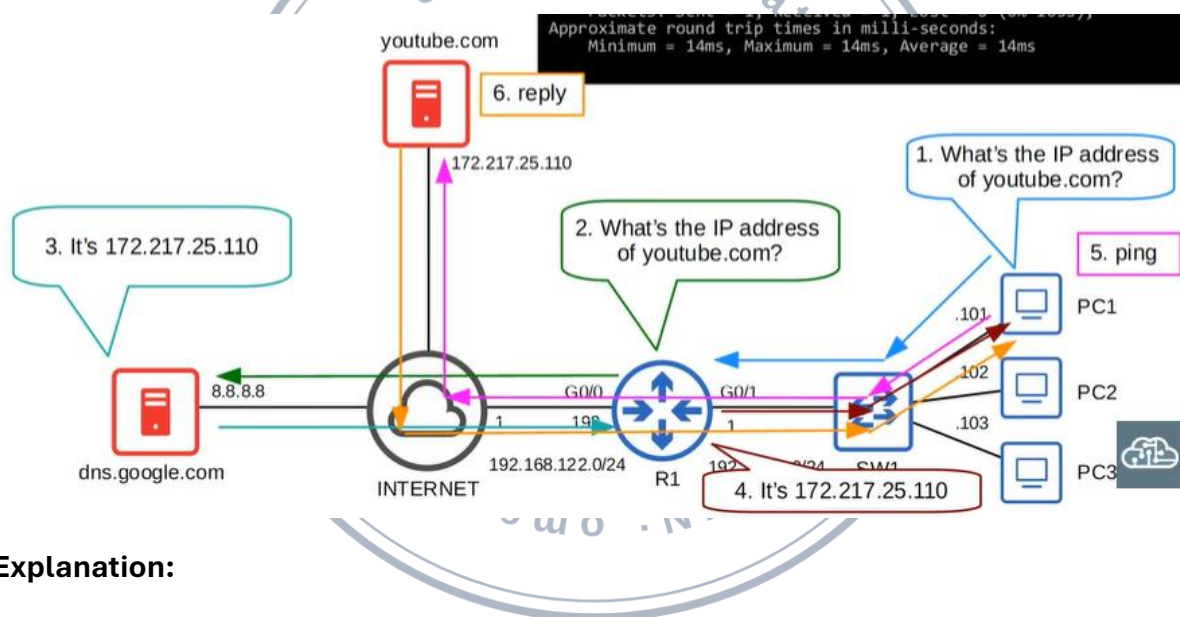
```
C:\Users\user>ping youtube.com -n 1

Pinging youtube.com [172.217.25.110] with 32 bytes of data:
Reply from 172.217.25.110: bytes=32 time=14ms TTL=117

Ping statistics for 192.168.0.102:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 14ms, Average = 14ms
```

Flow Diagram:

1. PC1 → R1 "What is youtube.com?"
2. R1 → 8.8.8.8 "What is youtube.com?"
3. 8.8.8.8 → R1 "172.217.25.110"
4. R1 → PC "172.217.25.110"
5. PC1 → YouTube (Ping)
6. YouTube → PC1 (Reply)



Explanation:

- PC1 queries R1
- R1 has **no local entry**
- R1 acts as **DNS client** and queries Google DNS
- Google replies with IP
- R1 forwards result to PC1

This shows dual behaviour:

- R1 = DNS server (for PC1)
- R1 = DNS client (to Internet)

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Viewing DNS Entries on Router

CLI Output of R1:

```
R1#show hosts
Default domain is not set
Name/address lookup uses domain service
Name servers are 8.8.8.8

Codes: UN - unknown, EX - expired, OK - OK, ?? - revalidate
       temp - temporary, perm - permanent
       NA - Not Applicable None - Not defined

Host                Port  Flags      Age Type  Address(es)
youtube.com         None (temp, OK)  0  IP    172.217.25.110
R1                  None (perm, OK)  4  IP    192.168.0.1
PC1                 None (perm, OK)  1  IP    192.168.0.101
PC2                 None (perm, OK)  4  IP    192.168.0.102
PC3                 None (perm, OK)  4  IP    192.168.0.103
```

Explanation:

This command shows both types of entries:

- **Permanent (perm)** → manually configured (ip host)
- **Temporary (temp)** → learned from DNS server

Temporary entries can expire and must be relearned.

Router as DNS Client Only

How to configure a Cisco router as a DNS client Basically, two of the command I already showed you are used to configure a router to act as a DNS client.

So, to demonstrate this I deleted all DNS settings from R1.

Then I tried to ping youtube.com but it didn't work.

CLI Output:

```
R1(config)#do ping youtube.com
Translating "youtube.com"
% Unrecognized host or address, or protocol not running.
```

```
R1(config)#ip name-server 8.8.8.8
```

Configure R1 to use the specified DNS server.

```
R1(config)#ip domain lookup
```

Enable R1 to perform DNS queries. (default)

```
R1(config)#do ping youtube.com
Translating "youtube.com"...domain server (8.8.8.8) [OK]

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.217.25.110, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/13 ms
```

```
R1(config)#ip domain name jeremysitlab.com
```

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Explanation:

- Initially, R1 cannot resolve the name
- After configuration:
 - It queries DNS server
 - Name resolution works

Important:

- As DNS client → works for R1 itself
- Not a DNS server → will not answer PC queries

Default Domain Name Configuration (Optional)

CLI Output: `R1(config)#ip domain name jeremysitlab.com`

Explanation: This sets a default domain name for the router.

Effect: Short names are automatically expanded

Example: ping PC1 → ping PC1.jeremysitlab.com

Notes:

- Older version: ip domain-name
- Required for features like SSH (covered later)

Command Summary

Windows:

```
C:\Users\user>ipconfig /all
C:\Users\user>nslookup name
C:\Users\user>ipconfig /displaydns
C:\Users\user>ipconfig /flushdns
C:\Users\user>ping ip-address -n number
```

Cisco IOS:

```
R1(config)#ip dns server
R1(config)#ip host hostname ip-address
R1(config)#ip name-server ip-address
R1(config)#ip domain lookup
R1(config)#ip domain name domain-name
R1#show hosts
```